

CSC 309

Data Structure and Algorithm

Examination: Semester Examination	Semester: Second Semester
Year: 2022	Duration: 2 hours
Total Questions: 5	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) Define recursion and iteration. Compare their advantages and disadvantages. (7 marks)
- (b) Write a recursive function to calculate the Fibonacci sequence. Trace the execution for $n=5$. (8 marks)

Question 2

Explain the following graph traversal algorithms with examples: (a) Breadth-First Search (BFS) (b) Depth-First Search (DFS). State their applications. (15 marks)

— End of Examination —